Test Report

Report No. CFT-2024031102BEN

Sample Name: Oriented Strand Boards (OSB)

Applicant: Jiangsu Space-O Technology Group Co., Ltd.

Test Type: Type Test

CFT Inspection and Testing Center

National Forestry and Grassland National Forest Reserves

Engineering Technology Research Center

Precautions

1. Test report is invalid if not affixed with authorized stamp of test or paging seal.
2. The test report is invalid without the corresponding signatures.
3. Test report is invalid only if being supplemented, deleted or altered.
4. Unless otherwise stated, the results shown in this test report are responsible for the delivered sample(s).
5. Objections to the test report must be submitted to CFT within 15 days of report received date.
6. The test applicant is responsible for authenticity of sample information which not subject to verification of CFT.

Lab Add: No. 48 Hongjin Road, Yubei District, Chongqing, People's Republic of China
Tel: 86-23-67277634 **E-mail:** admin@cft-es.com

Part 1: Sample and Test Information

Applicant	:	Jiangsu Space-O Technology Group Co., Ltd.
Address	:	No.9 Fukang Road, Xieqiao Town, Jingjiang City, Jiangsu Province 214500, China
Sample description	:	Oriented Strand Boards (OSB) samples for Type Test in this report were randomly sampled at the facility site by official representative of CFT Corporation in accordance with BS EN 300: 2006 on March 4, 2024. The samples were drawn from 3 production batches. The nominal thickness of OSB is 12 mm. Sampled specimen quantity, specimen specification and specimen source distribution are shown in below Table:

Sampling Plan				
Test Item		Specimen Specification (Length×Width× Thickness, mm)	Sample Size	Sample Distribution and Origin Distribution
General				
Dimensions		2440×1220×12	10	Sampled from 3 production batches. Sampling of panels shall be in accordance with EN 326-1.
Moisture Content		50×50×12	10	Sampled from 3 production batches. For each test specimen weighing at least 20 g, and the specimens shall be full thickness of the sheet.
Density		50×50×12	10	Sampled from 3 production batches. Sampling and cutting of the test pieces shall be carried out in accordance with EN 326-1.
Formaldehyde emission		25×25×12	N.A.	Sampled from 3 production batches. The test pieces are to be taken evenly distributed, over the width of the board but excluding a 500 mm wide strip at either end of the board. Perforation extraction is repeated for two times, each group is about 110g all thickness test pieces.
Mechanical and swelling properties				
Bending strength	major axis	290×50×12	30	Sampled from 3 production batches. Sampling and cutting of the test pieces shall be carried out according to EN 326-1. The longitudinal sample is consistent as the major axis of the particle.
	minor axis	290×50×12	30	Sampled from 3 production batches. Sampling and cutting of the test pieces shall be carried out according to EN 326-1. The transverse sample is consistent as the minor axis of the particle.

Internal bond	50×50×12	30	Sampled from 3 production batches. Sampling and cutting of the test pieces shall be carried out according to EN 326-1.
Swelling in thickness-24h immersion	50×50×12	30	Sampled from 3 production batches. Sampling and cutting of the test pieces shall be carried out according to EN 326-1.
Moisture resistance			
Moisture resistance	50×50×12	10	Sampled from 3 production batches. Sampling and cutting of test pieces per panel shall be carried out according to the principles of EN 326-1.

Adhesive Information : The OSB is consist of three layers whose wood direction are perpendicular to each other. The inner layer is made of Polymethylene Polyphenylene Isocyanate wherein phenolic formaldehyde resin face and back layer. Adhesive information is listed in the table below:

Adhesive Name	Type	Manufacturer
Polymethylene Polyphenylene Isocyanate	WANNATE -9132FC	Wanhua Chemical Group Co., Ltd.
Phenolic Formaldehyde Resin	13M319	AICA (Guangdong) Co., Ltd.

Manufacturer : Hubei Baoyuan Wood Industry Co., Ltd.

Manufacturer Address : No.18, Ziling Street, Ziling Town, Dongbao District, Hubei Province, China

Country of origin : P.R.C.

CFT Sample No. : 2024031102B1-B30

Sample Received Date : March 11, 2024

Testing Period : March 11, 2024 ～ March 29, 2024

Test Requested : Required type test(s) as per BS EN 300: 2006

Test Results : Please refer to next page(s).

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Part 2: Test Results Summary

Test Item		Units	Test Results[1]	Standard Requirements[2]	Verification Results
General requirements					
Tolerances on nominal dimensions	Thickness	mm	0.4	±0.8mm	Pass
	Length	mm	2.0	±3.0mm	Pass
	Width	mm	2.0	±3.0mm	Pass
Edge straightness tolerance		mm/m	0.9	1.5mm/m	Pass
Squareness tolerance		mm/m	0.6	2.0mm/m	Pass
Moisture Content		%	5.6	2%~12%	Pass
Tolerance on the mean density within a board		%	13.6	±15%	Pass
Formaldehyde Emissions		mg/100g	0.21	≤8mg/100g	E1
Requirements for specified mechanical and swelling properties					
Bending strength-major axis		N/mm²	24.9	Oriented Strand Boards shall comply with the requirements about load-bearing boards for use in humid conditions (Type OSB/3) listed in Table 4 of BS EN 300: 2006.	Pass
Modulus of elasticity in bending-major axis		N/mm²	4300		Pass
Bending strength-minor axis		N/mm²	10.2		Pass
Modulus of elasticity in bending-minor axis		N/mm²	1500		Pass
Internal bond		N/mm²	0.38		Pass
Swelling in thickness-24 h immersion		%	14.0		Pass
Requirements for moisture resistance					
Internal bond after boil test		N/mm²	0.18	≥0.13	Pass

Note:

(1) For detailed test data, please reference appendix I of test report.

(2) The standard requirements of specified mechanical, swelling properties and moisture resistance listed in Table 1, 4 and 5 of BS EN 300: 2006.

Compile: Lee wenqiang

Verify: Jean Tong

Approve: Jay Lee

Testing agency (official stamp):

Issue Date: April 2, 2024

CFT CORPORATION Inspection & Testing Services 5001141305077

Test Results

EN 322-1993—Moisture content

Test Method : Reference to section 6 moisture content determination stated in EN 322-1993.

Test Item(s)	Limit	Unit	Test Results
Moisture content	2～12	%	5.6

Note: The average value is used for the test results.

EN 323-1993—Tolerance on the mean density

Test Method : Reference to section 6 density determination stated in EN 323-1993.

Test Item(s)	Limit	Unit	Test Results
Tolerance on the mean density	≤15	%	13.6

Note: The average value is used for the test results.

EN 317-1993—24h thickness swelling

Test Method : Reference to section 6 thickness swelling determination stated in EN 317-1993.

Test Item(s)	Limit	Unit	Test Results
24h Thickness swelling	≤15	%	14.0

Note: The 95% percentile value of single sample shall be calculated according to section 6 in EN 326-1-1994.

EN 319-1993—Internal bond

Test Method : Reference to section 6 internal bond determination stated in EN 317-1993.

Test Item(s)	Limit	Unit	Test Results
Internal bond	≥0.32	N/mm^2	0.38

Note: The lower 5%-quantile value $Lq_{5\%}$ of single sample shall be calculated according to formula (6a) in section 7.3.6 of EN 326-1-1994.

EN 1087-1-1995—Moisture resistance

Test Method : Reference to section 6 internal bond determination stated in EN 1087-1-1995.

Test Item(s)	Limit	Unit	Test Results
Moisture resistance	≥0.13	N/mm^2	0.18

Note:

(1) The lower 5%-quantile value $Lq_{5\%}$ of single sample shall be calculated according to formula (6a) in section 7.3.6 of EN 326-1-1994.

(2) The test results meet the requirements of option 2 of table 5 in BS EN 300: 2006: internal bond strength after boil test.

EN 310-1993—Bending strength、Modulus of elasticity in bending

Test Method : Reference to section 6 bending strength and bending modulus of elasticity determination (three-point bending) stated in EN 310-1993.

Test Item(s)	Limit	Unit	Test Results
Bending strength-major axis	≥20	N/mm^2	24.9
Modulus of elasticity in bending-major axis	≥3500	N/mm^2	4300
Bending strength-minor axis	≥10	N/mm^2	10.2
Modulus of elasticity in bending-minor axis	≥1400	N/mm^2	1500

Note: The lower 5%-quantile value $Lq_{5\%}$ of single sample shall be calculated according to formula (6a) in section 7.3.6 of EN 326-1-1994.

BS EN ISO 12460-5-2015—Formaldehyde emission

Test method: Reference to formaldehyde emission determination stated in BS EN ISO 12460-5-2015.

Test Item(s)	Limit	Unit	Test Results	Corrected Value
Formaldehyde emission	≤ 8	mg/100g	0.21	0.26

Note:

(1) The samples are not subjected to conditioning treatment as per request by applicant.

(2) The formaldehyde emission are adjusted to the standard state value of 4.7% moisture content.

Appendix I - Test Data

1. Moisture Content

Dimension Specifications (mm)	Sample No.	Initial Mass (g)	Oven-Dried Mass (g)	Moisture Content (%)
50×50×12	1	17.13	16.29	5.2%
	2	18.88	17.97	5.1%
	3	18.82	17.24	9.2%
	4	19.46	18.52	5.1%
	5	18.46	17.54	5.2%
	6	17.37	16.47	5.5%
	7	17.76	16.81	5.7%
	8	17.28	16.32	5.9%
	9	17.80	17.04	4.5%
	10	20.75	19.86	4.5%

2. Density

Dimension Specifications (mm)	Sample No.	Length (mm)	Width (mm)	Thickness (mm)	Mass (g)	Density (kg/m^3)
50×50×12	1	50.56	50.84	11.950	17.13	557.67
	2	50.81	50.94	11.959	18.88	609.96
	3	50.89	50.51	11.957	18.82	612.33
	4	50.38	50.65	11.923	19.46	639.62
	5	50.80	50.61	11.945	18.46	601.10
	6	49.77	50.89	11.949	17.37	573.94
	7	50.39	50.81	11.919	17.76	581.98
	8	50.84	50.77	12.007	17.28	557.57
	9	50.93	49.77	11.925	17.80	588.87
	10	50.58	50.74	11.919	19.75	645.65

3. Formaldehyde emission

Sample specifications (mm)	Items	Blank test	Sample test	
25×25×12	Absorbance A	0.014	0.022	0.023
	Gradient of the calibration curve	12.8312		
	Mass of test pieces (g)		111.39	110.28
	Moisture content (%)		4.67	
	Perforator value (mg/100g)		0.19	0.22
	Average perforation value (mg/100g)		0.21	
	Moisture content factor F		1.24	
	Corrected perforation value		0.26	

4. Bending Strength and Modulus of Elasticity

Dimension Specifications (mm)	Sample No.	Width (mm)	Thickness (mm)	Limit Load (N)	Span (mm)	(P/△) (N/mm)	MOR (N/mm²)	MOE (N/mm²)
290×50×12 -major axis	1	50.67	12.10	620.0	240	114.5	30.1	4410.0
	2	50.73	12.12	720.0	240	153.7	34.8	5880.0
	3	50.55	12.08	570.0	240	133.3	27.8	5170.0
	4	50.74	12.08	910.0	240	151.7	44.2	5860.0
	5	50.46	12.12	690.0	240	144.0	33.5	5540.0
	6	50.24	12.18	620.0	240	122.1	29.9	4650.0
	7	50.77	12.10	730.0	240	133.8	35.4	5140.0
	8	50.63	12.08	480.0	240	113.9	23.4	4410.0
	9	50.71	12.05	790.0	240	146.1	38.6	5690.0
	10	50.60	12.11	800.0	240	144.8	38.8	5570.0
	11	50.63	12.02	810.0	240	158.0	39.9	6210.0
	12	50.42	12.04	800.0	240	140.3	39.4	5510.0
	13	50.36	12.04	680.0	240	123.9	33.5	4870.0
	14	50.24	12.10	580.0	240	114.3	28.4	4440.0
	15	50.40	12.12	710.0	240	120.0	34.5	4620.0
	16	50.62	12.06	870.0	240	144.6	42.5	5630.0
	17	50.22	12.03	630.0	240	129.0	31.2	5100.0
	18	50.36	12.05	730.0	240	139.0	35.9	5450.0
	19	50.25	12.03	790.0	240	135.9	39.1	5370.0
	20	50.45	12.05	640.0	240	129.8	31.5	5080.0
	21	50.54	12.05	850.0	240	154.5	41.7	6040.0
	22	50.39	12.09	550.0	240	117.5	26.9	4560.0
	23	50.28	12.07	740.0	240	135.1	36.4	5280.0
	24	50.47	12.07	510.0	240	111.2	25.0	4330.0
	25	50.68	12.06	600.0	240	128.6	29.3	5000.0
	26	50.70	12.07	900.0	240	139.6	43.9	5410.0
	27	50.62	12.05	970.0	240	147.4	47.5	5750.0
	28	50.33	12.06	810.0	240	146.1	39.8	5720.0
	29	50.74	12.12	730.0	240	155.8	35.3	5960.0
	30	50.38	12.06	700.0	240	132.2	34.4	5170.0

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Dimension Specifications (mm)	Sample No.	Width (mm)	Thickness (mm)	Limit Load (N)	Span (mm)	(P/△) (N/mm)	MOR (N/mm^2)	MOE (N/mm^2)
290×50×12 -minor axis	1	50.36	12.06	310.0	240	45.8	15.2	1790.0
	2	50.33	12.06	370.0	240	56.2	18.2	2200.0
	3	50.23	12.05	270.0	240	47.0	13.3	1850.0
	4	50.25	12.14	280.0	240	47.6	13.6	1830.0
	5	50.36	12.05	290.0	240	43.9	14.3	1720.0
	6	50.44	12.02	300.0	240	43.1	14.8	1700.0
	7	50.25	12.08	300.0	240	49.7	14.7	1940.0
	8	50.41	12.05	350.0	240	45.9	17.2	1800.0
	9	50.51	12.03	300.0	240	50.4	14.8	1980.0
	10	50.65	12.08	250.0	240	48.1	12.2	1860.0
	11	50.34	12.04	230.0	240	43.2	11.3	1700.0
	12	50.25	12.03	260.0	240	46.8	12.9	1850.0
	13	50.35	12.06	260.0	240	45.7	12.8	1790.0
	14	50.21	12.05	260.0	240	40.2	12.8	1580.0
	15	50.55	12.02	270.0	240	42.2	13.3	1660.0
	16	50.24	12.06	340.0	240	41.6	16.8	1630.0
	17	50.32	12.00	290.0	240	43.8	14.4	1740.0
	18	50.20	12.02	290.0	240	44.6	14.4	1770.0
	19	50.30	12.06	270.0	240	48.8	13.3	1910.0
	20	50.50	12.05	370.0	240	50.1	18.2	1960.0
	21	50.42	12.08	210.0	240	42.7	10.3	1660.0
	22	50.63	12.05	310.0	240	49.0	15.2	1910.0
	23	50.52	12.05	300.0	240	43.7	14.7	1710.0
	24	50.36	12.05	270.0	240	41.0	13.3	1610.0
	25	50.74	12.06	420.0	240	62.3	20.5	2420.0
	26	50.44	12.05	230.0	240	37.3	11.3	1460.0
	27	50.36	12.04	290.0	240	50.1	14.3	1970.0
	28	50.47	12.04	310.0	240	48.4	15.3	1900.0
	29	50.63	12.06	320.0	240	51.4	15.6	2000.0
	30	50.34	12.02	460.0	240	54.4	22.8	2150.0

Notes: P/△ is the linear elastic slope of the load-displacement graph.

5. Internal bond

Dimension Specifications (mm)	Sample No.	Length (mm)	Width (mm)	Limit Load (N)	Internal bond (N/mm^2)
50×50×12	1	50.71	49.69	1461.5	0.58
	2	50.07	49.87	1647.9	0.66
	3	50.76	49.62	1209.0	0.48
	4	50.44	49.23	1762.9	0.71
	5	50.38	50.16	1010.8	0.40
	6	50.31	50.00	1358.5	0.54
	7	50.13	49.78	2021.4	0.81
	8	50.93	49.12	1375.9	0.55
	9	50.24	50.10	1636.1	0.65
	10	50.12	49.83	1673.4	0.67
	11	50.38	49.58	1373.8	0.55
	12	50.49	49.14	1290.2	0.52
	13	50.00	49.94	1173.6	0.47
	14	50.51	49.71	1707.2	0.68
	15	50.06	49.53	1165.4	0.47
	16	50.95	49.26	1531.2	0.61
	17	50.79	49.11	1571.4	0.63
	18	50.57	49.45	1500.4	0.60
	19	50.67	49.46	1603.9	0.64
	20	50.78	49.40	1455.1	0.58
	21	50.10	49.92	1826.0	0.73
	22	50.08	49.96	1776.6	0.71
	23	50.18	50.02	1556.2	0.62
	24	50.84	49.42	1532.5	0.61
	25	50.06	49.29	912.9	0.37
	26	50.45	49.68	1553.7	0.62
	27	50.78	49.05	747.3	0.30
	28	50.93	49.12	1025.6	0.41
	29	50.98	48.67	1513.6	0.61
	30	50.10	50.15	1432.0	0.57

6. Swelling in thickness

Dimension Specifications (mm)	Sample No.	Thickness before test (mm)	Thickness after test (mm)	Swelling in thickness (%)
50×50×12	1	12.093	13.486	11.5%
	2	12.117	13.451	11.0%
	3	12.111	13.588	12.2%
	4	12.082	13.144	8.8%
	5	12.019	13.408	11.6%
	6	12.055	13.002	7.9%
	7	12.067	13.280	10.1%
	8	12.099	13.361	10.4%
	9	12.091	13.337	10.3%
	10	12.056	13.476	11.8%
	11	12.069	13.269	9.9%
	12	12.155	13.355	9.9%
	13	12.059	13.426	11.3%
	14	12.088	13.413	11.0%
	15	12.119	13.801	13.9%
	16	12.077	13.196	9.3%
	17	12.111	13.287	9.7%
	18	12.073	13.328	10.4%
	19	12.074	13.411	11.1%
	20	12.062	14.064	16.6%
	21	12.096	13.474	11.4%
	22	12.028	13.667	13.6%
	23	12.102	13.445	11.1%
	24	11.990	13.237	10.4%
	25	12.049	13.464	11.7%
	26	12.090	13.191	9.1%
	27	12.112	13.303	9.8%
	28	12.110	13.643	12.7%
	29	12.128	13.562	11.8%
	30	12.144	13.716	12.9%

7. Moisture resistance-Internal bond after boil test

Dimension Specifications (mm)	Sample No.	Length (mm)	Width (mm)	Limit Load (N)	Internal bond (N/mm^2)
50×50×12	1	50.50	50.79	554.8	0.22
	2	50.67	50.87	763.3	0.30
	3	50.04	50.82	621.9	0.24
	4	50.14	50.70	750.3	0.30
	5	50.97	50.76	733.0	0.28
	6	51.00	50.49	674.6	0.26
	7	50.82	50.53	622.9	0.24
	8	50.58	50.97	679.8	0.26
	9	50.48	50.41	551.9	0.22
	10	50.49	50.07	749.8	0.30
	11	50.57	50.38	561.8	0.22
	12	50.96	50.49	500.4	0.19
	13	50.64	50.70	789.2	0.31
	14	50.94	50.14	669.7	0.26
	15	50.05	50.66	520.1	0.21
	16	50.21	50.58	740.5	0.29
	17	50.43	50.47	684.0	0.27
	18	50.30	50.31	559.2	0.22
	19	50.92	50.09	509.2	0.20
	20	50.44	50.34	598.9	0.24
	21	50.37	50.09	553.2	0.22
	22	50.74	50.94	547.0	0.21
	23	50.40	50.07	664.6	0.26
	24	50.91	50.55	529.1	0.21
	25	50.06	50.68	531.5	0.21
	26	50.09	50.10	575.9	0.23
	27	50.43	50.86	528.0	0.21
	28	50.82	50.98	631.7	0.24
	29	50.82	50.80	525.1	0.20
	30	50.20	50.63	670.9	0.26

Representative Sample Photos

CFT authenticate the photo on original report only.

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